

Cem Ozcelik

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PROFESSIONAL SUMMARY

Senior Data Scientist with an Industrial Engineering background and 4+ years translating complex datasets into actionable strategy insights across manufacturing, supply chain, and financial domains. Advanced SQL and Python practitioner with hands-on experience delving into structured and unstructured data -- ETL pipeline engineering, deep EDA, KPI definition, and interactive analytical dashboard development. Industrial Engineering training delivers a native instinct for business-first KPI design: before building a metric, I ask what decision it enables and what it costs to measure wrong. Comfortable delivering findings to both technical teams and non-technical management. Bilingual Turkish-English.

CORE COMPETENCIES

Skills: Advanced SQL, KPI Management & Definition, EDA & Pattern Discovery, Statistical Analysis, Business Intelligence, Stakeholder Reporting, Data Storytelling

Tools: Python, SQL (PostgreSQL / MSSQL / Oracle), PySpark / Databricks, Django, Docker, Scikit-learn, XGBoost

WORK EXPERIENCE

Anadolu Pet Akvaryum & Effeffe Jan 2026 - Present

Senior Database Analyst | Maltepe, Istanbul

- Designed and maintained automated reporting systems and dashboards centralizing operational visibility and supporting key business decisions across supply chain and inventory management.
- Defined and managed route-level KPI benchmarks -- cost-to-profit ratios per delivery route -- delivering a **3% improvement in overall logistics efficiency**.
- Engineered advanced supply chain analytics in Python to drive demand forecasting, sustaining an **8-hour weekly reduction in operational overtime**.
- Translated SKU-level Association Rule Mining outputs into strategic cross-sell recommendations, driving revenue growth through data-informed merchandising decisions.
- Tech:** Python, Django, REST APIs, PostgreSQL, Docker, Association Rule Mining, SQL

Anadolu Pet Akvaryum & Effeffe Sept 2024 - Dec 2025

IT & Data Analytics Specialist | Maltepe, Istanbul

- Built the company's data analytics infrastructure from scratch -- a Django web application for centralized reporting, dashboard delivery, and performance metrics tracking across departments.
- Conducted Route-Level Profitability Analysis using advanced SQL across Oracle ERP data, identifying high-risk logistics segments and establishing cost-to-profit KPI standards.
- Engineered predictive supply chain optimization models in Python and PL/SQL, reducing stockouts and improving inventory turnover.
- Tech:** Python, Django, PL/SQL, Oracle, SOLID, full-stack deployment

Birinci Automotive Jan 2022 - Sept 2024

Mid-Level Data Analyst | Cayirova, Kocaeli

- Delved into years of high-frequency machine sensor data to uncover hidden OEE improvement opportunities -- mapping failure signatures, identifying candidate features, and building performance dashboards aligning production metrics with manufacturing KPIs.
- Developed a Predictive Maintenance & Anomaly Detection framework using unsupervised clustering on historical failure patterns, delivering a **15% weekly increase in critical machine availability** and **15% OEE improvement**.
- Engineered robust ETL pipelines in MSSQL and Python to ingest structured manufacturing data, replacing manual Excel workflows with sustainable analytical infrastructure.
- Tech:** Python, MSSQL, ETL, unsupervised clustering, time-series EDA

Junior Data Analyst | Cayirova, Kocaeli

- Conducted EDA and applied statistical analysis methods to workforce datasets, uncovering hidden patterns to optimize HR policies and improve operational planning.
- Designed automated data workflows and ingestion pipelines, improving reporting efficiency by 15%.
- **Tech:** Python, SQL, statistical analysis, ETL

PROJECTS

End-to-End Credit Risk Prediction -- EDA Platform & ML github.com/ccemozclk/Credit-Risk

Interactive analytical platform for credit default risk detection -- Django EDA dashboard enabling non-technical stakeholders to explore data and interrogate model decisions. XGBoost optimized for heavy class imbalance (**63% F1-Score**, temporal holdout, zero leakage), fully containerized via Docker. Architecture transfers directly to customer analytics: swap credit features for ARPU or churn signals, same platform.

Python, XGBoost, Django, Docker, PostgreSQL, Scikit-learn

Scalable Recommendation Engine -- Large-Scale Behavioral Analytics github.com/ccemozclk/Spark-Recommendation-Engine

End-to-end behavioral segmentation and personalized recommendation system on Apache Spark -- processing large-scale unstructured datasets at distributed scale on Databricks. K-Means for customer segmentation; ALS collaborative filtering for personalized item-level recommendations tuned to business KPI (conversion-rate uplift), not just RMSE.

PySpark, Databricks, K-Means, ALS collaborative filtering

AI-Powered Medical Diagnostics -- Explainable AI & CNNs github.com/ccemozclk/Medical-Diagnostics

End-to-end diagnostic support system for pneumonia detection from chest X-rays, optimizing for Recall (FN cost >> FP cost). **97% Recall** via Transfer Learning (VGG16, ResNet50) + Grad-CAM heatmaps for clinical explainability and bias detection.

TensorFlow-Keras, Transfer Learning, Grad-CAM, class imbalance handling

EDUCATION

B.Sc. Industrial Engineering -- Kirikkale University

Sept 2016 - Feb 2021

Rigorous foundation in statistical analysis, optimization (MILP with IBM ILOG CPLEX), and multi-criteria decision-making (AHP, TOPSIS). Cost-function thinking and KPI design applied directly to every analytical project.

CERTIFICATIONS

Associate Data Scientist Career Track -- [DataCamp](#) (PySpark, Feature Engineering, Recommendation Engines)

Jan - May 2024

Data Science & Machine Learning Bootcamp -- [Miuul](#) (6 months, 10+ real-world projects)

Mar - Sept 2022

SKILLS

Languages: Python (Advanced), SQL (Advanced), Bash, JavaScript (Basics)

Analytics & ML: Scikit-learn, XGBoost, TensorFlow, PyTorch, Pandas, NumPy, Statistical Analysis, A/B Testing

Data Engineering: PostgreSQL, MSSQL, Oracle, ETL Pipelines, PL/SQL, Apache Spark (PySpark), Databricks

Deployment & Web: Docker, Django, FastAPI, RESTful APIs, Git, Linux

Human Languages: Turkish (Native), English (Fluent), Spanish (Basic)